

EC400: Mathematical Methods for Macroeconomics

This part of the course introduces students to the methods of dynamic optimization and the systems of dynamic equations. These methods are widely used in many areas of macroeconomics — including growth theory, business cycles, monetary economics, search models, etc. — but are also common in other fields of economics. The main goal of the course is to provide mathematical *tools* that can be used to solve dynamic problems. We will focus mostly on applications and heuristic derivations.

Course outline:

Lecture 1: basic consumption-savings problem, direct approach, Euler equation, transversality condition, continuous time, Hamiltonian, Maximum Principle

Lecture 2: difference and differential equations, states and controls, dynamic systems, steady state, stability, phase diagrams, Blanchard-Kahn method, linearization

Lecture 3: backward induction, value function, Bellman equation, Principle of Optimality, HJB equation, reinforcement learning

References: the lecture notes for the course are self-contained, but the following books are useful to study the methods introduced in this course:

- Blanchard and Fischer, *Lectures on macroeconomics*, MIT press, 1989.
- Galor, *Discrete Dynamical Systems*, Springer Science & Business Media, 2007.
- Ljungqvist and Sargent, *Recursive Macroeconomic Theory*, MIT Press, 2004.
- Romer, *Macroeconomic Theory*, University of California, Berkeley, 2018.
- Stokey, *The Economics of Inaction*, Princeton University Press, 2008.
- Stokey, Lucas and Prescott, *Recursive Methods in Economic Dynamics*, Harvard University Press, 1989.
- Sutton and Barto, *Reinforcement learning: An introduction*, MIT press, 2018.